# Atharva Vidwans

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Ph.D. student making quantum algorithms practical by reducing the shots, queries, and circuit resources they need, with work spanning Bayesian amplitude estimation, Hamiltonian simulation, gate set tomography, and fault-tolerant resource estimation. Seeking a quantum computing research internship for Summer 2027.

## Education

Ph.D. Chemistry, University of Wisconsin-Madison (expected May 2028) *Sep '24 – Present*
M.S. Physics (Quantum Computing), University of Wisconsin-Madison *Aug '22 – May '24*
B.E. Mechanical Engineering, Savitribai Phule Pune University *Jul '15 – Jun '19*

## Publications

- **Atharva Vidwans** et al. "Not every circuit deserves a shot: adaptive shot experiment design for gate set tomography." Manuscript in preparation.
- *(co-first author)* Jingcheng Dai* and **Atharva Vidwans*** et al. "Molecular resonance identification in complex absorbing potentials via integrated quantum computing and high-throughput computing." arXiv preprint arXiv:2511.15981, 2025.
- *(co-first author)* Qilin Li* and **Atharva Vidwans*** et al. "Harnessing Bayesian Statistics to Accelerate Iterative Quantum Amplitude Estimation." *Quantum* **10**, 1962 (2026). arXiv:2507.23074.
- **Atharva Vidwans**. "Cognitive computing in autonomous vehicles." In *Cognitive Computing for Human-Robot Interaction*, pp. 121–146. Academic Press, 2021.

## Work Experience

**Argonne National Laboratory**, Givens Associate (summer fellowship) *May '26 – Aug '26*
Advisors: Dr. Matt Menickelly and Dr. Jeffrey Larson

- Reduced required shots in gate set tomography relative to the standard Levenberg-Marquardt baseline by about 2.1x (one qubit) and 2.7x (two qubits) at the same gate accuracy, with no loss in SPAM accuracy, by allocating shots adaptively across circuits.
- Developed the allocation method using Fisher-information (D-optimal) experiment design, and designed a new criterion that fixes its SPAM-accuracy loss.
- Made the allocator 3.4x faster with an exact algorithm, making two-qubit runs faster on HTCondor (CHTC) with pyGSTi and POUNDERS (a derivative-free optimizer).
- Selected for the 2026 FASTMath Student Cohort (DOE SciDAC Institute) on mentor recommendation.

## Research Experience

**Bayesian and Resource-Estimation Methods for Double-Bracket Quantum Imaginary-Time Evolution** *Jan '25 – Present*
Advisor: Prof. Matthew Otten, UW-Madison

- Reduced queries per energy estimate by 4.2x (LiH) and 9.8x ($BeH_2$) in noiseless simulation, and a further 1.7 to 2.2x with an improved generator, using a Bayesian measurement scheme that reuses the failed ancilla outcomes PDBQITE throws away and picks each measurement to maximize information gain.
- Cut state-preparation cost by 4.1x (LiH) and 137x ($BeH_2$) with a recovery step that undoes a failed PDBQITE step instead of restarting, without changing the final state.
- Built fault-tolerant resource estimates with a surface-code wall-clock model (FLASQ) and double-factorized Trotter circuits compiled to Clifford+T (ffsim, Qiskit, WISQ), and ran systems up to $Fe_2S_2$ on HTCondor (CHTC).

**Bayesian Iterative Quantum Amplitude Estimation (BIQAE)** *Jan '23 – Jun '25*
Advisors: Prof. Micheline Soley and Prof. Yazhen Wang, UW-Madison

- Reduced sample complexity by about 14% compared to IQAE across six orders of magnitude in target accuracy, and by 8 to 13% for ground-state energies of $H_2$, LiH, $BeH_2$, and HF (co-first author, *Quantum* 2026).
- Built a unified statistical framework for amplitude estimation and added Bayesian inference to Iterative QAE, with rigorous intervals at every iteration.
- Showed with proofs and simulations that Bayesian inference alone accounts for the speedup.

**qDRIVE for Molecular Resonances with Complex Absorbing Potentials** *Jan '23 – Jun '25*
Advisor: Prof. Micheline Soley, UW-Madison

- Developed qDRIVE (co-first author), which combines the complex absorbing potential formalism with deflation-based VQE to find molecular resonance energies and wavefunctions.
- Ran the VQE task network asynchronously in parallel with HTCondor DAGMan to cut wall time.
- Recovered bound and resonance energies of a benchmark model to under 1% error in statevector simulation, and tested under shot noise, an IBM Torino noise model, and IBM hardware.

**Surface-Code Logical Qubit Simulation** *Feb '23 – May '23*
Advisor: Prof. Shimon Kolkowitz, UW-Madison

- Built X-cut and Z-cut surface-code logical qubits in Qiskit with logical X, Y, Z, and H gates, and started a braiding-based logical CNOT.
- Compared logical and physical qubits under gate and measurement noise up to 2%, and the logical qubits had 2 to 4.5x lower infidelity at the highest noise level.

**Quantum Optimization for Vehicle Routing** *Jun '21 – May '22*
Advisor: Prof. Pawel Gora, University of Warsaw

- Formulated the vehicle routing problem as a QUBO and implemented Layer-VQE in Qiskit, which grows the ansatz one identity-initialized layer at a time. Early experiments showed it outperforming standard VQE.
- Worked on Filtering VQE, a combined Layer-Filtering VQE, and CVaR objectives to improve accuracy.

## Technical Skills

- **Quantum:** gate set tomography (pyGSTi), VQE, amplitude estimation (IQAE/BIQAE), Hamiltonian simulation, Trotterization, double factorization, surface codes, fault-tolerant resource estimation, Qiskit, ffsim, WISQ, OpenFermion, PySCF.
- **Statistics and optimization:** optimal experiment design, Fisher information, Bayesian statistics, derivative-free optimization (POUNDERS).
- **Computing:** Python, NumPy, SciPy, Numba, PyCUDA, HTCondor, Docker, Git, LaTeX.

## Presentations

- "Efficient gate set tomography with POUNDERS." Talk, Argonne National Laboratory, Jul 2026.
- "Quantum Deflated Resonance Identification Variational Eigensolver (qDRIVE) Algorithm for Molecular Resonances on Near-Term Quantum Computers." **A. Vidwans**, J. Dai, E. Wan. Talk, Chicago Quantum Summit, 2024.
- "qDRIVE: A Variational Quantum Eigensolver for Resonance Identification on Near-Term Computers." Talk, Wisconsin Quantum Institute Colloquium, Mar 2024.
- "Quantum Deflated Resonance Identification Variational Eigensolver Algorithm for Molecular Resonances on Near-Term Quantum Computers." **A. Vidwans**, J. Dai. Poster, Graduate Student Faculty Liaison Committee, UW-Madison, Aug 2023.
- "Solving Vehicle Routing Problem Using Variational Quantum Eigensolver." **A. Vidwans**, W. el Maouaki. Talk, 13th Warsaw IT Days, Poland, Apr 2022.

## Other Experience

**Chegg India**, Subject Matter Expert, Advanced Mathematics (part time) *Feb '21 – May '22*
**Comofi Medtech Pvt. Ltd.**, Medical Imaging Intern *Feb '20 – Jul '20*
Advisor: Mr. Rakesh Sharma

- Reduced DRR generation time from 70 min to 8 sec for CT-to-X-ray registration in robotic kidney stone surgery, by writing a ray-casting algorithm in Python and parallelizing it with Numba. A PyCUDA kernel ran about 200x faster than Numba CUDA.
- Registered intra-operative C-arm X-rays against DRRs using MSE, SSIM, NCC, and feature-based homography to track kidney calyx shift for the robotic arm.

## Awards, Service and Teaching

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| IBM Certified Associate Developer, Quantum Computation using Qiskit v0.2X | *Jan '22* |
| Rank 4 globally, IBM Qiskit Challenge 2021 | *Jun '21* |
| Fourth position, National Level Qiskit Challenge, India (QWorld) | *Sep '20* |
| Ad hoc reviewer, *Quantum Machine Intelligence* (Springer) | *Jun '23* |
| Teaching Assistant, General Chemistry (CHEM 104, 109) and Physical Chemistry 2 (562) | *Sep '24 – Dec '25* |